Lab2

FILE : [LAB2.pkt]

นายเกียรติศักดิ์ นันทรัตน์ 673380072-9 ทำ part1-3 configure ให้กับ switch, router, pc-a, pc-b, pc-c
นายนครินทร์ ณ ระนอง 673380274-7 Section.2
นายศุภวัฒน์ ข่ายทอง 673380062-2 Section.2 : Part 4 และ worksheet 1 VERIFICATION COMMANDS CHECKLIST (Router , Switch)
นายภาคิน เมฆสุวรรณ 673380054-1 Section2:Worksheet 2: Design Understanding\
นายสรวิชญ์ วันเสน 673380064-8 Section.2 :Part 5,VERIFICATION COMMANDS CHECKLIST(PCs)

📝 LAB DELIVERABLES

Worksheet 1: Configuration Check

Checkpoint	Command	Expected	Actual	Pass/Fail
VLANs exist	show vlan brief	VLANs 10,20,99	S1#show vlan brief VLAN Name Status Ports ---- ---------- --------- ------- 1 default active Gig0/2 10 USERS active Fa0/1 20 SERVERS active Fa0/2 99 MANAGEMENT active Fa0/3 999 BLACKHOLE active Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24 1002 fddi-default active 1003 token-ring-default active 1004 fddinet-default active 1005 trnet-default active	Pass
Trunk active	show int trunk	VLANs allowed	S1#show interfaces trunk Port Mode Encapsulation Status Native vlan Gig0/1 on 802.1q trunking 1 Port Vlans allowed on trunk Gig0/1 10,20,99 Port Vlans allowed and active in management domain Gig0/1 10,20,99 Port Vlans in spanning tree forwarding state and not pruned Gig0/1 10,20,99	Pass
Router subifs	show ip int br	up/up	R1#show ip interface brief Interface IP-Address OK? Method Status Protocol GigabitEthernet0/0 unassigned YES unset up up GigabitEthernet0/0.10 192.168.10.1 YES manual up up GigabitEthernet0/0.20 192.168.10.65 YES manual up up GigabitEthernet0/0.99 192.168.10.129 YES manual up up GigabitEthernet0/1 unassigned YES unset administratively down down GigabitEthernet0/2 unassigned YES unset administratively down down Vlan1 unassigned YES unset administratively down down	Pass
Inter-VLAN ping	ping	Success	PC-A C:\>ping 192.168.10.10 Pinging 192.168.10.10 with 32 bytes of data: Reply from 192.168.10.10: bytes=32 time<1ms TTL=128 Reply from 192.168.10.10: bytes=32 time<1ms TTL=128 Reply from 192.168.10.10: bytes=32 time=5ms TTL=128 Reply from 192.168.10.10: bytes=32 time=9ms TTL=128 Ping statistics for 192.168.10.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 9ms, Average = 3ms	Pass

Checkpoint	Command	Expected	Actual	Pass/Fail
			PC_C C:\>ping 192.168.10.70 Pinging 192.168.10.70 with 32 bytes of data: Reply from 192.168.10.70: bytes=32 time<1ms TTL=127 Reply from 192.168.10.70: bytes=32 time<1ms TTL=127 Reply from 192.168.10.70: bytes=32 time<1ms TTL=127 Reply from 192.168.10.70: bytes=32 time<1ms TTL=127 Ping statistics for 192.168.10.70: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms	

Worksheet 2: Design Understanding

1.Why is VLAN 99 used for management?

เพราะอยากแยกการจัดการอุปกรณ์เครือข่ายออกจากการใช้งานปกติ จะได้ปลอดภัยขึ้นและไม่ปะปนกับข้อมูลผู้ใช้

2.Why is VLAN 999 assigned to unused ports?

เพื่อความปลอดภัย ถ้ามีใครมาเสียบสายมั่ว ๆ จะไม่สามารถเข้าเครือข่ายหลักได้

3.What does 802.1Q tagging do?

คือการติดป้ายบอกว่าเฟรมข้อมูลนี้อยู่ VLAN ไหน ทำให้หลาย VLAN วิ่งผ่านสายเดียวกันได้

4.Why does each VLAN require a different subnet?

เพราะแต่ละ VLAN แยกกันเหมือนเป็นคนละเครือข่าย ต้องมี subnet ต่างกันถึงจะจัดการและส่งข้อมูลได้ถูกต้อง

🔍 VERIFICATION COMMANDS CHECKLIST

Router

- show run

```
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.192
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.10.65 255.255.255.192
!
interface GigabitEthernet0/0.99
 encapsulation dot1Q 99
 ip address 192.168.10.129 255.255.255.192
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end
```

```
R1#show run
Building configuration...

Current configuration : 979 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX15246137-
!
!
!
!
!
!
!
!
!
!
!
!
!
```

- show ip route

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 6 subnets, 2 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.10.64/26 is directly connected, GigabitEthernet0/0.20
L       192.168.10.65/32 is directly connected, GigabitEthernet0/0.20
C       192.168.10.128/26 is directly connected, GigabitEthernet0/0.99
L       192.168.10.129/32 is directly connected, GigabitEthernet0/0.99
```

- show arp

```
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.10.70            9   0003.E4B4.2BCB  ARPA   GigabitEthernet0/0.20
Internet  192.168.10.130           9   0004.9A6B.AE86  ARPA   GigabitEthernet0/0.99
```

Switch

- show vlan brief

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

- show interfaces trunk

```
S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,99
```

- show mac address-table

```
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0050.0f05.2a01    DYNAMIC     Gig0/1
  10    0050.0f05.2a01    DYNAMIC     Gig0/1
  20    0050.0f05.2a01    DYNAMIC     Gig0/1
  99    0050.0f05.2a01    DYNAMIC     Gig0/1
```

PCs

PC1

- Ipconfig

```
C:\>Ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2E0:F7FF:FEC2:B678
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.65

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

- arp -a

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.1          0050.0f05.2a01        dynamic
  192.168.10.65         0050.0f05.2a01        dynamic
```

- Ping

PC-A Ping หา PC-B

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time=1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC-A Ping หา PC-C

```
C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC2

- Ipconfig

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::203:E4FF:FEB4:2BCB
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.70
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.65

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

- arp -a

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.65         0050.0f05.2a01        dynamic
```

- ping

PC-B Ping หา PC-A

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC-B Ping หา PC-C

PC3

- Ipconfig

```
C:\>Ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::204:9AFF:FE6B:AE86
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.130
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.129

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

- arp -a

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.129        0050.0f05.2a01        dynamic
```

- ping

PC-C Ping หา PC-A

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC-C Ping หา PC-B

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

🎓 ASSESSMENT QUESTIONS

1. Which device performs inter-VLAN routing?
   Answer. Router 0 เพราะเป็นตัวที่มี sub-interface สำหรับแต่ละ VLAN
2. Why can PCs in different VLANs communicate?
   Answer. เพราะ Router 0 ทำ Inter-VLAN routing ผ่าน trunk link ทำให้PC-a , PC-B , PC_C ที่อยู่คนละ Vlan ส่งข้อมูลหากันได้
3. What breaks if the trunk is removed?
   Answer. การสื่อสารระหว่าง VLAN จะหยุดทำงาน Router จะไม่รับ VLAN Tag
4. Why is VLAN 1 avoided?
   Answer. เพราะเป็น Default VLAN และใช้สำหรับ Control Traffic
   เสี่ยงต่อปัญหาด้านความปลอดภัย
5. What security risk exists if unused ports remain active?
   Answer. คนอื่นสามารถเสียบอุปกรณ์เข้าพอร์ตที่ไม่ได้ใช้งานได้ อาจเข้าถึงเครือข่ายโดยไม่ได้รับอนุญาต